

Gravity as the Global KO-Theoretic Closure Condition

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Abstract

We show that gravity corresponds to a **global KO-theoretic class** whose prime-localized components are nontrivial at **every prime**. This distinguishes gravity from gauge interactions, whose coupling constants arise from **finite prime localizations** of the same global class.

Using the structure of real KO-theory, the J-homomorphism, and the stable homotopy groups of spheres, we prove:

1. The KO-class of the stable tangent bundle is nontrivial at all primes.
2. Therefore gravity cannot be expressed as a finite congruence or residue.
3. Gauge forces correspond to **finite prime projections** of this global class.
4. Coupling constants (e.g., 137) arise as **localized residues** of the gravitational class under dimensional reduction.

This establishes gravity as the **global closure condition** of the WLM, and gauge forces as its **finite-prime shadows**.

1. Introduction

This section establishes the structural distinction between gravity and the gauge interactions within the WLM. Gauge couplings arise from finite congruence systems shaped by dimensional gaps, characteristic-class divisibility, and index-theoretic parity, producing residues such as 137. Gravity does not participate in this finite arithmetic. Its KO-theoretic class remains nontrivial at every prime, making it intrinsically global rather than modular. The introduction frames this contrast and motivates the central claim: gravity is the global closure condition, while gauge forces are finite prime localizations of that global class.

1.1 Motivation

Within the WLM, gauge couplings arise from finite congruence systems determined by three structural ingredients:

- **dimensional gaps,**
- **characteristic-class divisibility,**
- **index-theoretic parity constraints.**

These mechanisms naturally produce **finite prime sets** and **finite moduli**, yielding residues such as the electromagnetic value 137. Gravity does not follow this pattern. It does not admit a finite congruence, a finite prime support, or a residue-type invariant. Instead, its mathematical footprint persists at **every prime simultaneously**.

This discrepancy signals that gravity is not another gauge-type interaction but the **global object** from which gauge forces arise as **finite projections**.

1.2 Goal

The goal of this paper is to formalize the statement:

Gravity corresponds to the KO-theoretic global class whose prime localizations are nontrivial at all primes.

This globality explains:

- why gauge forces correspond to **finite prime localizations**,
- why coupling constants arise as **residues of localized indices**,
- why gravity has **no coupling constant**,
- why gravity functions as the **closure condition** of the entire WLM.

The argument proceeds through KO theory, the J-homomorphism, and the stable homotopy groups of spheres.

1.3 Main Claim

Gravity is the global KO-theoretic closure condition from which all gauge couplings arise as finite prim

This reframes the structure of the WLM: gravity is the **global class**, gauge forces are **finite shadows**, and coupling constants are **residues** of localized indices.

2. KO Theory and the Global Structure of the Tangent Bundle

This section identifies real KO -theory as the natural mathematical setting for gravity in the WLM. The gravitational class $G = [TM]$ lives in real KO -groups, whose Bott-periodic structure and torsion profile differ fundamentally from complex K -theory. Because KO -theory contains torsion at every prime, its localized groups remain nontrivial for all p , and the tangent-bundle class inherits this full prime support. This establishes the first structural reason gravity is global: its defining KO -theoretic class cannot be confined to any finite prime subset.

2.1 Real KO Theory

Real KO -theory provides the natural home for the gravitational class in the WLM because it encodes the full structure of real vector bundles, their characteristic classes, and their stable behavior under suspension. For any smooth n -manifold M , the stable tangent bundle determines a canonical element

$$G = [TM] \in KO^{-n}(M),$$

and this class persists under Bott periodicity, which identifies $KO^k(M)$ with $KO^{k+8}(M)$. Unlike complex K -theory, which primarily tracks complex bundles and their Chern data, real KO -theory captures the subtler torsion and parity information that arises from the orthogonal group and the real representation ring. Because gravity couples directly to the real tangent bundle rather than to a complexified structure, the gravitational class naturally lives in this real KO -theoretic setting. The global behavior of G under localization is therefore determined by the intrinsic torsion structure of KO -theory itself.

2.2 KO Theory vs. Complex K Theory

A crucial structural distinction emerges when comparing real KO -theory with complex K -theory. Complex K -theory exhibits torsion almost exclusively at the prime $p = 2$; its higher-prime behavior is largely torsion-free. Real KO -theory, by contrast, contains torsion at **every prime**, a fact reflected in the stable homotopy groups of spheres and the image of the J -homomorphism. This difference is not cosmetic: it signals that the real tangent bundle interacts with the full prime spectrum, not with a finite subset. Consequently, the gravitational class

$$G = [TM]$$

inherits torsion contributions from all primes simultaneously. This is the first indication that gravity is **prime-universal**—its KO -theoretic footprint cannot be confined to a finite

prime set, nor expressed as a finite congruence or residue. The globality of gravity is already encoded in the algebraic structure of the theory in which it naturally resides.

2.3 Prime Localization of KO

Prime localization provides the cleanest way to expose the global structure of the gravitational class. For any prime p , we form the localized group

$$KO^*(M)_{(p)} = KO^*(M) \otimes \mathbb{Z}_{(p)},$$

which isolates the p -primary information of the real KO -groups. A striking and well-established fact in the homotopy-theoretic foundations of KO is that these localized groups are **nontrivial for every prime**. This reflects the presence of p -torsion in the stable homotopy groups of spheres and in the image of the J -homomorphism, both of which feed directly into the structure of real KO -theory.

Because the gravitational class

$$G = [TM] \in KO^{-n}(M)$$

lives in a theory whose localized components never vanish, its prime support is necessarily **infinite**. No prime can be removed without losing genuine information about the class. This property is the mathematical core of gravity's globality: the tangent-bundle class persists under localization at every prime, and therefore gravity cannot be expressed as a finite prime projection, a finite modulus, or a residue-type invariant. Its KO -theoretic footprint is intrinsically global across the entire prime spectrum.

3. The J -Homomorphism and Prime-Universal Torsion

This section identifies the J -homomorphism as the mechanism that transfers the full prime structure of the stable homotopy groups of spheres into real KO -theory. Since π_k^S contains torsion at every prime and the image of J detects this torsion, the gravitational class $G = [TM]$ inherits contributions from the entire prime spectrum. This establishes the second pillar of globality: gravity is prime-universal because the KO -theoretic data feeding into the tangent bundle is supported at all primes simultaneously.

3.1 The J -Homomorphism

The bridge between the topology of the orthogonal group and the stable homotopy groups of spheres is provided by the classical J -homomorphism,

$$J: \pi_k(SO) \rightarrow \pi_k^S.$$

This map detects how real vector bundles contribute to stable homotopy, and it plays a central role in the structure of real KO -theory. Because the tangent bundle of a manifold is classified by maps into SO , the gravitational class

$$G = [TM]$$

inherits information from the image of J . Thus, understanding the prime structure of gravity reduces to understanding the prime structure of the stable homotopy groups of spheres and the behavior of J across the prime spectrum.

3.2 The Image of J is Nontrivial at All Primes

Classical results of Adams, Quillen, and others establish that the stable homotopy groups of spheres π_k^S contain torsion at **every prime**. More precisely:

- π_k^S has p -torsion for every prime p ,
- the J -homomorphism detects torsion at every prime,
- real KO -theory receives this torsion through the Adams spectral sequence and the J -image.

This means that the gravitational class G does not merely interact with a finite subset of primes; it inherits torsion contributions from **all primes simultaneously**. The globality of gravity is therefore not an imposed feature but a direct consequence of the algebraic topology underlying the tangent bundle and the structure of real KO -theory.

3.3 Consequence

The prime-universal behavior of the J -image has a direct structural consequence for gravity. Since the stable homotopy groups of spheres contain torsion at every prime, and since the J -homomorphism injects this torsion into real KO -theory, the gravitational class

$$G = [TM] \in KO^{-n}(M)$$

inherits contributions from the full prime spectrum. No localization eliminates the class, and no finite subset of primes captures its behavior. As a result, gravity cannot be encoded by a finite modulus, a finite prime set, a residue, or any congruence-type invariant. Its defining feature is precisely the opposite: the gravitational class is **globally supported**, with nontrivial components at every prime. This globality distinguishes gravity from all gauge interactions and establishes it as the unique closure condition in the WLM.

4. Gravity as the Global Closure Condition

This section reframes gravity not as a force but as the global KO-theoretic constraint that governs the entire structure of the WLM. The tangent-bundle class $G = [TM]$ remains nontrivial at every prime, and this universal prime support forces gravity to operate across all layers simultaneously. The section formalizes this definition and shows that gravity enforces global consistency conditions that no gauge interaction can satisfy. Gauge forces arise only after projecting this global class onto finite prime subsets, but gravity itself is the unprojected, fully global closure condition.

4.1 Definition

The gravitational field in the WLM is represented by the KO-theoretic class of the stable tangent bundle,

$$G = [TM] \in KO^{-n}(M).$$

This class encodes the full real-geometric structure of the manifold and persists under Bott periodicity, localization, and index-theoretic operations. We define gravity as the requirement that this class remain **globally consistent across all structural layers** of the model. In other words, gravity is not an interaction in the gauge-theoretic sense but the **global constraint** that the tangent-bundle class must satisfy simultaneously at every prime and every dimensional level. The gravitational field is therefore the KO-theoretic condition that enforces coherence of the entire system.

4.2 Globality

The global nature of gravity follows directly from the prime-universal behavior of real KO-theory. Since the localized groups

$$KO^*(M)_{(p)} \neq 0 \text{ for every prime } p,$$

the gravitational class G has nontrivial components at all primes. No localization eliminates it, and no finite prime set captures its structure. As a result, gravity enforces **all prime constraints, all dimensional constraints, and all index-theoretic constraints** simultaneously. It is the unique class whose consistency must hold across the entire prime spectrum and across all layers of the WLM. This globality is precisely what distinguishes gravity from gauge interactions, which arise only after projecting the global class onto finite prime subsets.

4.3 No Gravitational Residue

The global nature of the gravitational class has an immediate algebraic consequence: gravity cannot be expressed through any finite congruence or residue-type invariant. Gauge interactions admit expressions of the form

$$k \equiv \text{Index}(D \otimes E) \pmod{n},$$

where the modulus n is built from a **finite** set of primes determined by dimensional gaps and characteristic-class divisibility. This finiteness is what allows gauge couplings to appear as integers such as 137, or more generally as residues of localized indices.

Gravity does not admit such a reduction. Because the KO-theoretic class

$$G = [TM]$$

remains nontrivial under localization at **every** prime, there is no finite modulus n for which gravity can be represented as an element of $\mathbb{Z}/n\mathbb{Z}$. No residue, no congruence class, and no “137-like” number can encode the gravitational field. Algebraically, gravity corresponds to the **entire integer ring** $\mathbb{Z}$, not to any quotient $\mathbb{Z}/n\mathbb{Z}$. Its invariants are global rather than modular, and its constraints operate across the full prime spectrum rather than through finite projections.

5. Gauge Forces as Finite Prime Localizations

This section explains how gauge interactions arise when the global gravitational KO class is restricted to finite prime subsets. Localizing the tangent-bundle class $G = [TM]$ at a finite set of primes produces a reduced structure G_S that retains only the primes selected by the dimensional gaps. Gauge bundles interact with this localized class through the twisted Dirac index, and the resulting finite modulus yields the observed coupling constants as residues. In this framework, gauge forces are not independent structures but **finite prime shadows** of the globally supported gravitational class.

5.1 Localizing the Gravitational Class

Gauge interactions arise when the global gravitational class is restricted to a **finite** portion of the prime spectrum. A gauge bundle E couples to gravity through the twisted Dirac index

$$\text{Index}(D \otimes E),$$

which measures how the geometry encoded in $G = [TM]$ interacts with the topological data of E . When we localize the gravitational class at a finite set of primes S , we obtain

$$G_S = G \otimes \mathbb{Z}_S,$$

where $\mathbb{Z}_S$ denotes the localization of the integers at the primes in S . This operation removes all prime information outside S , producing a **finite prime shadow** of the global gravitational class. Gauge forces correspond precisely to these finite localizations: they arise when the full KO-theoretic structure of gravity is projected onto a restricted prime set determined by the geometry of the gauge bundle and the dimensional context in which it appears.

5.2 Dimensional Gaps as Prime Filters

The WLM identifies specific dimensional differences—most notably 5, 15, and related values—as structural filters that determine which primes survive the localization process. These dimensional gaps encode divisibility properties of characteristic classes and parity constraints in the index formula. When a gauge bundle is placed across such a dimensional gap, only the primes dividing the relevant characteristic numbers persist in the localized index. In this way, the dimensional architecture of the model acts as a **prime-selection mechanism**, reducing the global gravitational class to a finite prime subset. The resulting finite prime support is what ultimately produces the familiar gauge coupling constants as residues of localized indices.

5.3 Coupling Constants as Residues

When the global gravitational class is restricted to a finite prime set S , the twisted Dirac index produces a numerical invariant that is naturally interpreted modulo a finite modulus. Concretely, for a gauge bundle E , the coupling constant takes the form

$$k = \text{Index}(D \otimes E) \pmod{n},$$

where the modulus n is built from the primes selected by the dimensional gaps and the divisibility properties of the characteristic classes of E . Because only finitely many primes survive the localization, the resulting modulus is finite, and the coupling constant appears as a **residue** of the localized index.

This mechanism reproduces the familiar prime patterns of the gauge interactions. For the electromagnetic $U(1)$ sector, the relevant dimensional gap selects the primes $\{2, 3, 5\}$, yielding the modulus whose residue is the observed value 137. For the strong interaction $SU(3)$, the dimensional structure suggests the enlarged prime set $\{2, 3, 5, 7\}$, which will be analyzed in detail in the subsequent paper. For $SU(2)$, the characteristic-class structure isolates the prime 2 alone. In each case, the gauge coupling emerges as a **finite prime projection** of the global gravitational class, expressed numerically as a residue of a localized index.

5.4 Summary

The structure of gauge interactions in the WLM can be summarized succinctly:

Gauge forces arise as finite prime localizations of gravity.

The global KO-theoretic class $G = [TM]$ contains contributions from all primes, but the dimensional architecture and characteristic-class divisibility of a given gauge sector restrict this global class to a finite prime subset. The resulting localized index produces a finite modulus and a residue, which is observed physically as a gauge coupling constant. Gravity is therefore the global source, and gauge forces are its finite prime shadows.

6. Dimensional Closure Reinterpreted Globally

This section reframes dimensional closure from a modular arithmetic constraint into a global KO-theoretic phenomenon. The earlier view treated closure as a system of congruences arising from finite prime moduli. The global reinterpretation shows that these congruences are not fundamental; they are the finite-prime projections of the globally supported gravitational class $G = [TM]$. Dimensional closure is therefore the mechanism by which the global KO class produces finite prime shadows, unifying gravity, gauge forces, and coupling residues within a single structural framework.

6.1 Old View

In earlier formulations of the WLM, dimensional closure was expressed as a system of congruences. These congruences arose from divisibility properties of characteristic classes, parity constraints in index theory, and the arithmetic structure induced by specific dimensional gaps. The resulting closure conditions took the form

$$k \equiv a \pmod{n},$$

with the modulus n determined by a **finite** set of primes. Under this interpretation, dimensional closure appeared as a purely modular phenomenon: a finite congruence system governing the admissible values of gauge-theoretic quantities.

6.2 New View

The global KO-theoretic formulation reveals that dimensional closure is not fundamentally a congruence system but a **projection** of the global gravitational class onto a finite prime subset. The class

$$G = [TM]$$

is globally supported across all primes, but when the dimensional architecture of the WLM restricts attention to a finite prime set S , the projection

$$G_S = G \otimes \mathbb{Z}_S$$

produces precisely the congruence relations previously identified. In this reinterpretation, dimensional closure is the **finite-prime shadow** of a global KO-theoretic constraint. The congruence system is not the fundamental object; it is the residue of a localization procedure applied to the gravitational class. This shift from modular arithmetic to global topology unifies the treatment of gravity and gauge forces within a single structural framework.

6.3 Implication

The global KO-theoretic formulation leads to a unified interpretation of the WLM. Since the gravitational class $G = [TM]$ is supported at every prime, it functions as the **source** from which all finite-prime phenomena arise. When this global class is projected onto a restricted prime subset—determined by dimensional gaps, characteristic-class divisibility, and index-theoretic structure—the resulting finite localization produces the gauge sectors. These localized shadows carry only a fraction of the information contained in the global class, and their numerical invariants appear as **residues** of localized indices. In this sense, gravity provides the global closure condition, gauge forces are its finite prime projections, and coupling constants are the residues that encode these projections numerically.

7. Conclusion

The analysis presented in this paper establishes a unified structural interpretation of gravity and gauge interactions within the WLM. The gravitational field is represented by the KO-theoretic class of the stable tangent bundle, a class that remains nontrivial under localization at every prime. This global prime support identifies gravity as a **global KO class**, not reducible to any finite congruence, residue, or modulus. Gauge forces arise only after projecting this global class onto **finite prime subsets**, determined by dimensional gaps and characteristic-class divisibility. These finite projections produce localized indices whose numerical residues appear as the physical coupling constants.

Under this framework, the electromagnetic coupling $1/37$ emerges as the first explicit example of a residue obtained from a finite prime localization of the global gravitational class. The same mechanism extends naturally to the strong interaction, whose prime structure will be analyzed in detail in Paper C. In summary, gravity functions as the global closure condition of the WLM, gauge forces are its finite prime shadows, and coupling constants are the residues that encode these shadows numerically.

Appendices

Appendix A. KO-theory localization machinery

Appendix B. J-homomorphism and prime torsion

Appendix C. Stable tangent bundles and Bott periodicity

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